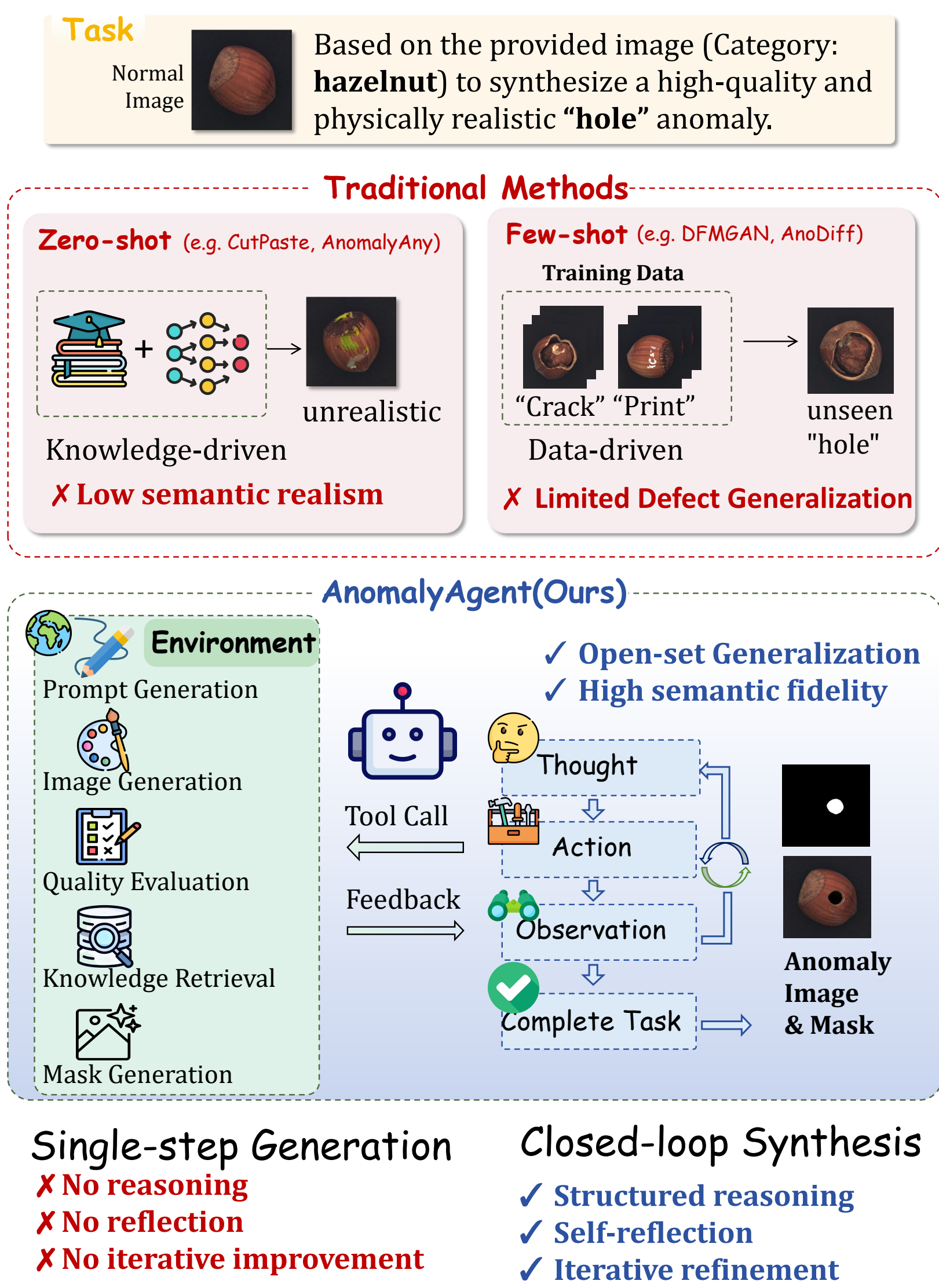
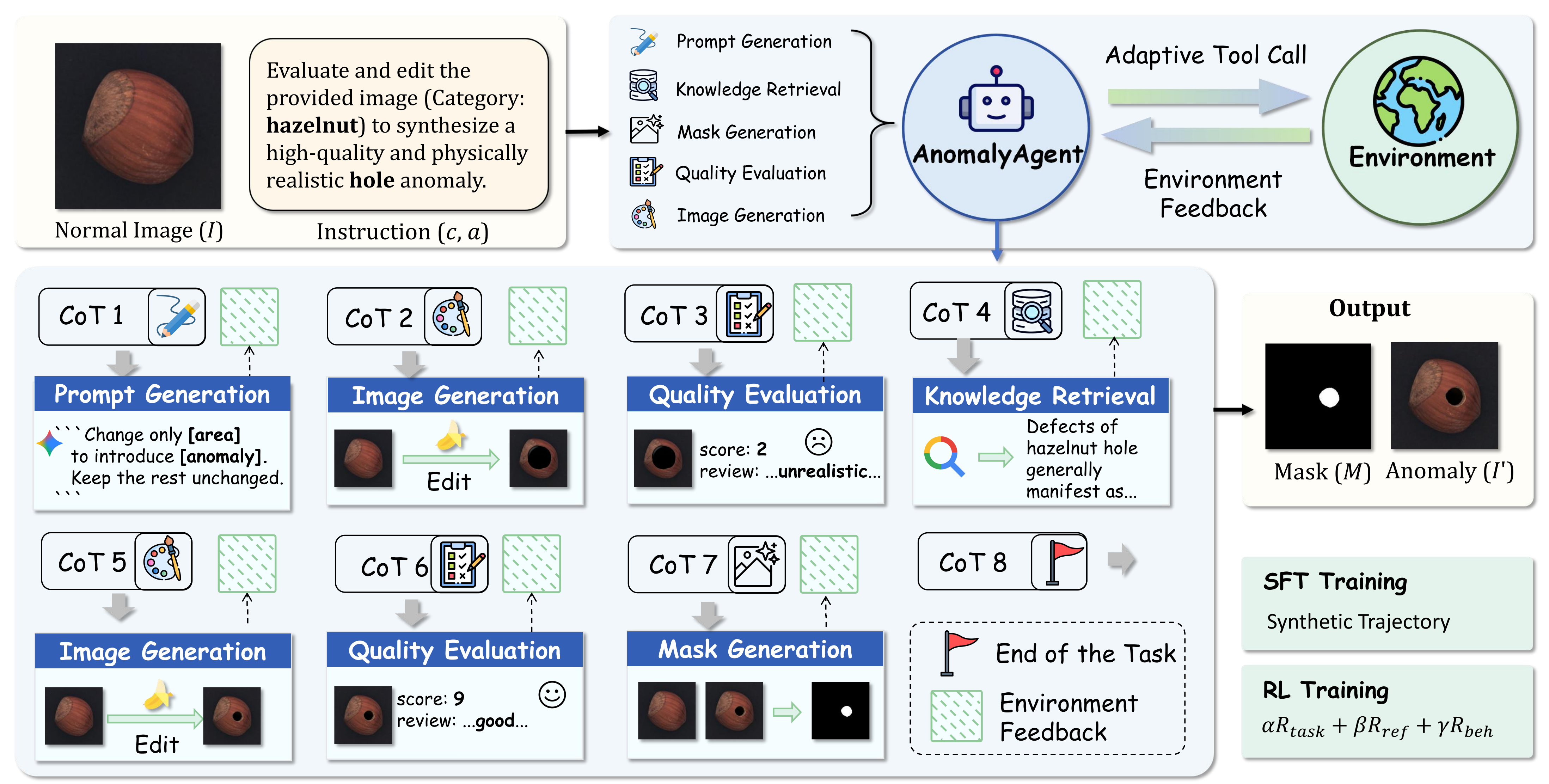


AnomalyAgent: Agentic Industrial Anomaly Synthesis via Tool-Augmented Reinforcement Learning

1. Introduction



2. Method



The challenge. Real defects are scarce and highly diverse.

Few-shot: limited defect coverage.

Zero-shot: often lack semantic realism.

Our idea. Use feedback to guide iterative synthesis and self-refinement.

Adaptive tool use. Given a normal image, its object category, and a target defect, the agent dynamically selects among five tools to generate an anomaly image and its pixel-level mask.

Closed-loop refinement. Quality scores and semantic feedback guide prompt revision, with knowledge retrieval when needed. Tool selection adapts to the current state, not a fixed sequence.

Two-stage training. SFT learns structured tool use from synthetic trajectories. GRPO then optimizes long-horizon decision-making with **task, reflection, and behavior rewards**.

3. Results

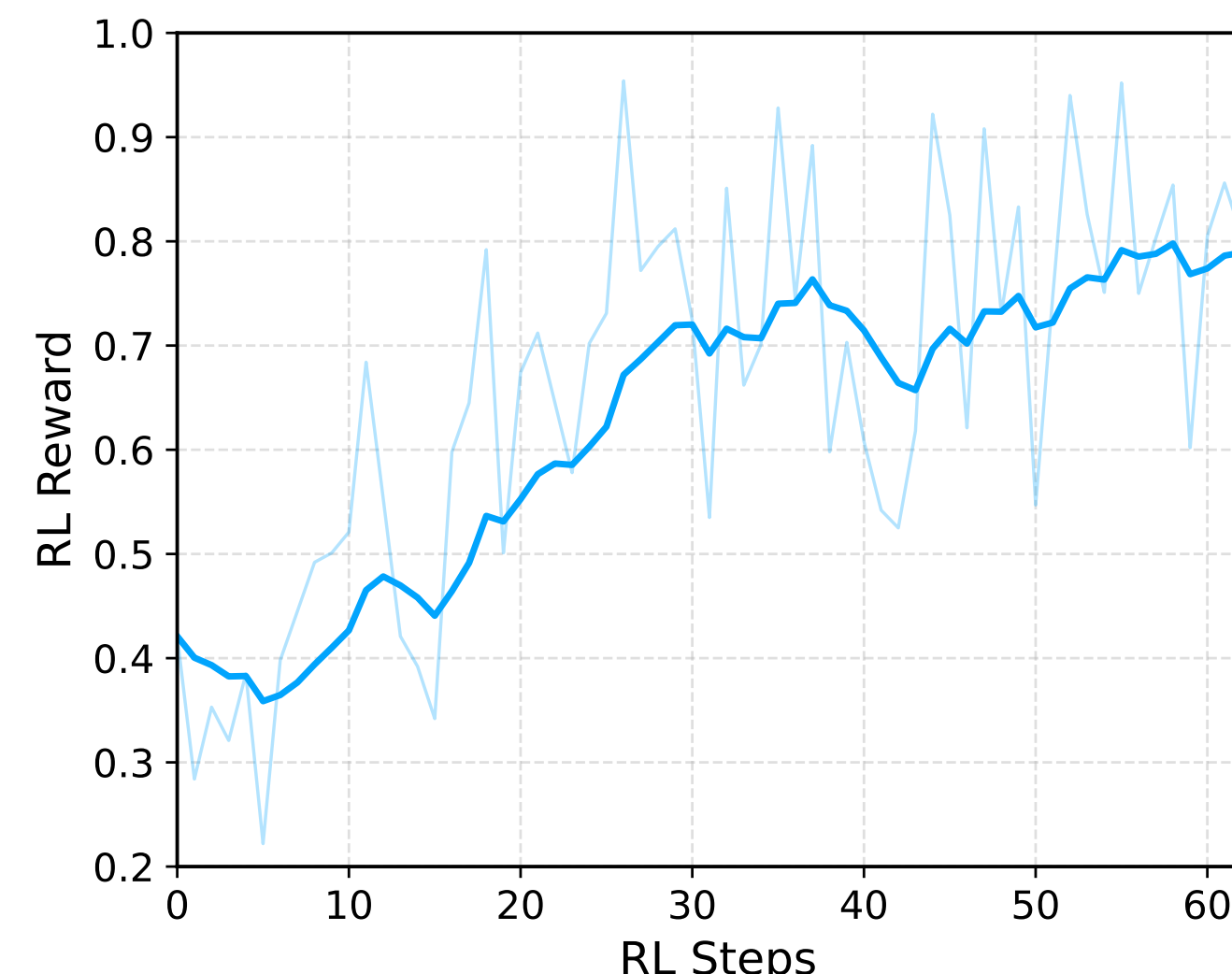
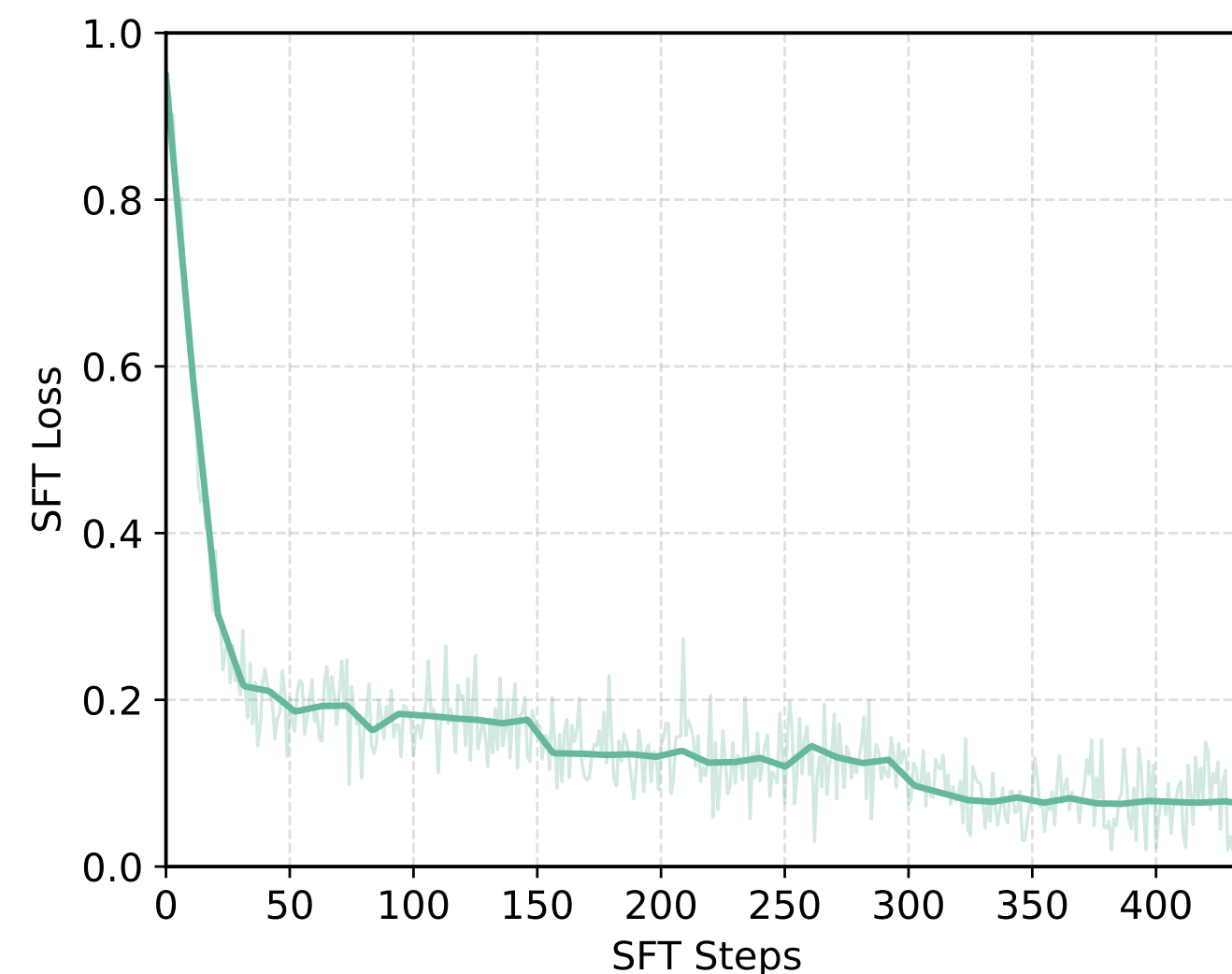
Category	AnoHybrid			Gemini-img			GPT-img			AnomalyAgent			AnoHybrid			Gemini-img			GPT-img			AnomalyAgent		
	Pixel-Level Performance									Image-Level Performance														
	AUC	AP	F ₁	AUC	AP	F ₁	AUC	AP	F ₁	AUC	AP	F ₁	AUC	AP	F ₁	AUC	AP	F ₁	AUC	AP	F ₁	AUC	AP	F ₁
bottle	98.3	77.2	72.5	96.1	74.5	68.1	86.0	52.7	51.5	97.2	73.5	69.7	99.2	99.8	98.7	98.7	99.5	97.7	99.5	99.8	97.7	98.7	99.6	98.1
cable	94.1	76.0	73.4	93.3	68.5	60.2	90.2	55.5	50.4	97.4	76.5	69.2	96.1	97.7	91.9	98.8	98.9	94.7	98.3	98.8	97.4	99.1	99.3	96.1
capsule	98.4	51.7	54.6	96.0	44.0	47.3	88.3	33.2	34.5	96.6	46.0	49.7	94.9	98.8	95.2	90.8	96.7	92.0	90.2	91.3	88.7	91.7	97.6	91.8
carpet	98.6	82.8	75.6	98.8	79.0	72.3	91.3	69.0	60.4	99.4	85.1	76.4	96.3	98.9	94.0	97.9	99.2	96.0	95.6	96.1	95.9	99.4	99.8	98.4
grid	98.8	58.6	59.2	97.4	27.5	38.6	89.0	22.1	32.7	97.8	40.3	44.9	100	100	100	95.6	98.0	93.7	95.7	98.0	93.7	100	100	100
hazelnut	99.6	89.4	82.6	96.7	81.1	77.4	90.3	78.3	70.2	99.4	86.4	80.1	96.7	98.3	92.6	96.2	97.0	90.4	94.1	97.9	96.0	98.9	99.2	95.9
leather	99.6	72.7	67.1	99.4	72.8	68.3	93.9	69.4	65.6	99.7	80.3	73.9	98.4	99.5	97.4	100	100	100	96.9	98.0	95.1	100	100	100
metal nut	98.8	93.5	87.0	97.5	90.6	83.6	86.2	58.5	58.6	99.3	95.4	88.5	99.8	99.9	99.2	99.4	99.8	97.7	95.5	98.6	93.5	99.8	99.9	98.7
pill	99.3	94.9	88.4	99.1	82.7	72.7	88.0	65.1	60.3	99.5	84.0	80.4	99.1	99.8	98.9	92.3	98.0	93.0	92.5	96.0	91.9	97.4	98.8	96.6
screw	77.0	7.8	6.4	97.6	10.5	17.3	94.7	21.6	29.9	98.4	55.4	57.8	44.6	72.6	84.9	88.4	94.5	88.1	90.9	95.3	88.6	94.6	97.8	91.3
tile	99.3	94.6	87.4	97.3	92.6	80.5	94.3	59.0	59.9	99.3	94.6	85.7	99.5	99.8	99.0	99.3	99.6	97.4	98.8	99.5	95.0	100	100	100
toothbrush	98.7	65.2	67.8	99.5	73.7	75.4	94.8	31.1	42.6	99.3	72.3	70.3	100	100	100	100	100	100	98.3	99.1	95.2	100	100	100
transistor	98.1	80.8	74.2	90.6	62.1	59.9	88.2	60.3	54.7	90.5	63.7	62.1	92.9	90.4	86.3	94.5	92.6	90.9	95.9	97.9	93.4	98.3	97.1	92.9
wood	95.8	70.7	64.8	95.8	78.5	72.8	82.2	45.6	49.0	97.5	79.8	74.0	96.6	98.7	98.7	98.5	99.3	97.7	97.3	98.0	94.3	100	100	100
zipper	99.1	82.3	74.9	96.9	75.8	66.4	90.4	55.0	49.4	98.9	79.8	72.4	99.9	100	99.3	99.2	99.7	97.0	96.3	98.1	94.2	99.9	100	99.4
Mean	96.9	72.9	69.1	96.8	67.6	64.1	89.9	51.8	51.3	98.0	74.2	70.3	94.3	96.9	95.7	96.6	98.2	95.1	95.7	97.5	94.0	98.5	99.3	97.3

Localization on MVTec-AD. Best mean performance across all six metrics.

98.0 pixel AUC / 98.5 image AUC. Pixel AP rises from 72.9 to **74.2** over AnoHybrid, providing more accurate spatial supervision for downstream detection.

Method	IS $\uparrow$	IC-L $\uparrow$
CutPaste	1.76	0.22
DRAEM	1.76	0.25
NSA	1.44	0.26
RealNet	1.64	0.22
AnomalyAny	2.02	0.33
AnoStyler	2.04	<u>0.32</u>
AnoHybrid	<u>2.06</u>	<u>0.32</u>
Gemini-img	1.91	0.29
GPT-img	1.77	0.29
Grok-img	1.68	0.28
AnomalyAgent	2.10	0.33

Method	Accuracy $\uparrow$
AnoStyler	32.2
AnoHybrid	<u>52.6</u>
Gemini-img	44.7
GPT-img	40.5
Grok-img	38.9
AnomalyAgent	57.0



Realistic defects. Precise masks.

Iterative tool-driven refinement reduces semantic drift and incoherent defect boundaries.

Generation quality

2.10 IS; joint-best 0.33 IC-L.

Classification

57.0% accuracy; +4.4 points over AnoHybrid.

Training dynamics

SFT loss converges; GRPO reward rises from **0.42 to 0.79**.

4. Conclusion

Anomaly synthesis as multi-turn decision-making.

AnomalyAgent integrates **structured tool use** and **iterative reasoning** for controllable, semantically consistent anomaly generation beyond single-step pipelines. **SFT on synthetic trajectories** teaches tool-use strategies, while **agentic RL** optimizes long-horizon decisions. Experiments demonstrate **improved generation quality** and **stronger downstream classification and localization** over existing zero-shot methods. Future work will scale to more complex industrial scenarios and broader multimodal generation tasks.